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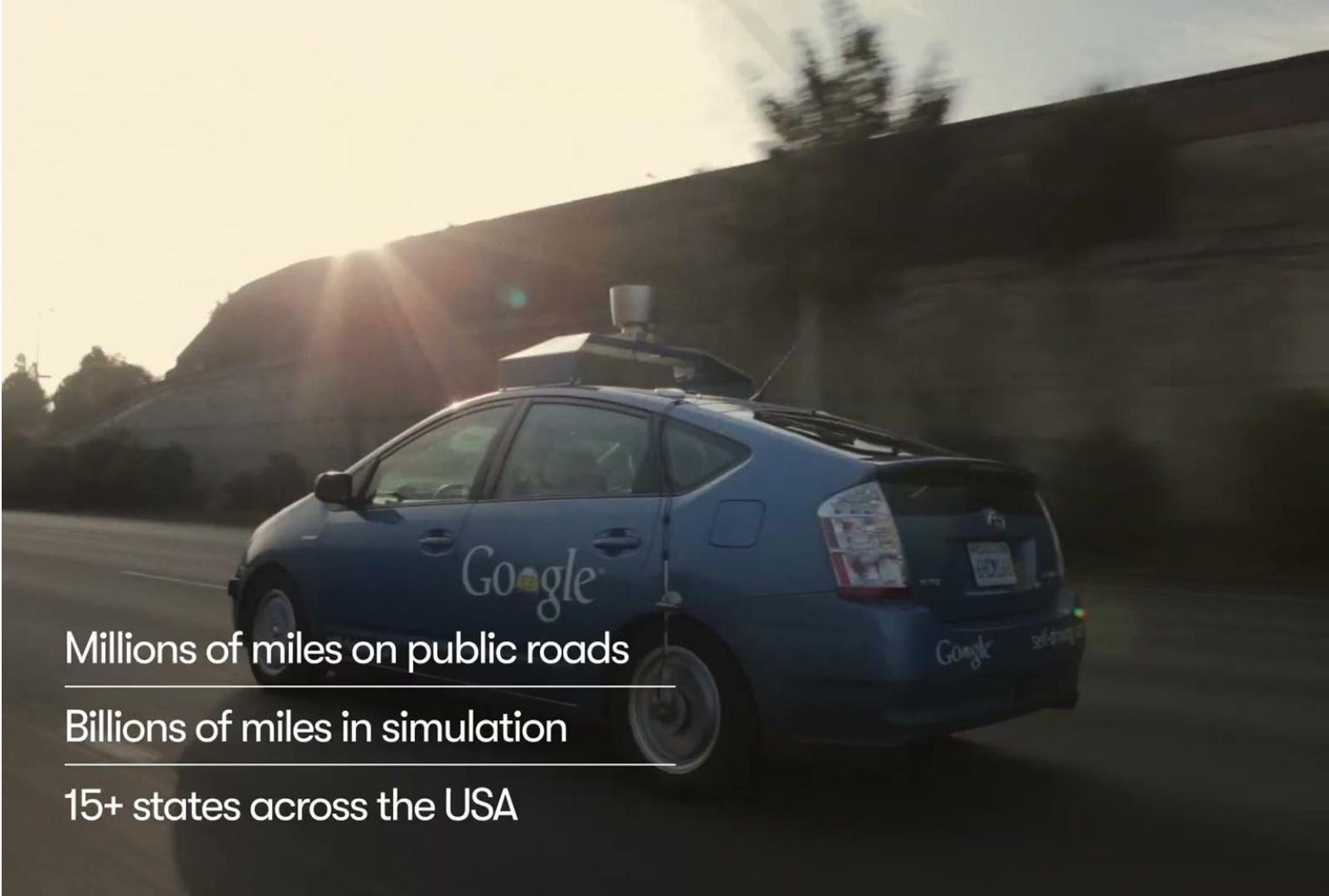
Computational human driver models for Automated Driving System (ADS) evaluation based on (en)active inference

Johan Engström

10th Summer School and Symposium on Computational Interaction (S³CIX), at the University of Glasgow, in Scotland
June 18th, 2026



Our experience



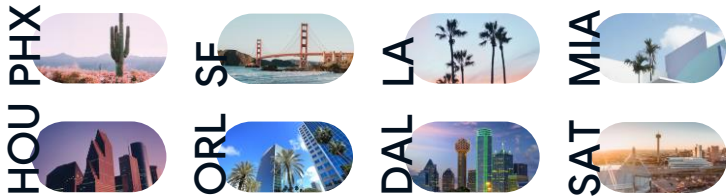
Millions of miles on public roads

Billions of miles in simulation

15+ states across the USA

Serving riders in —

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WAYMO | Uber

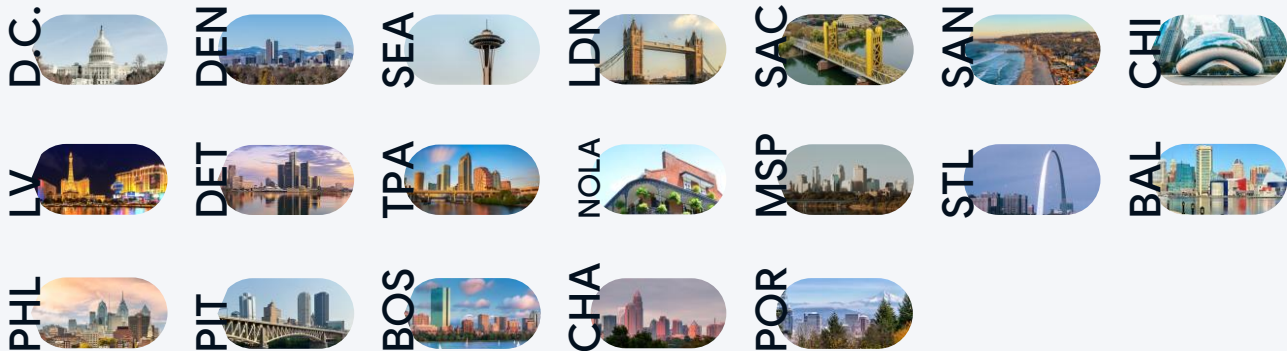


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Coming soon —

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Common question in ADS development: What would a human have done in this situation?



Need for computational human road user models

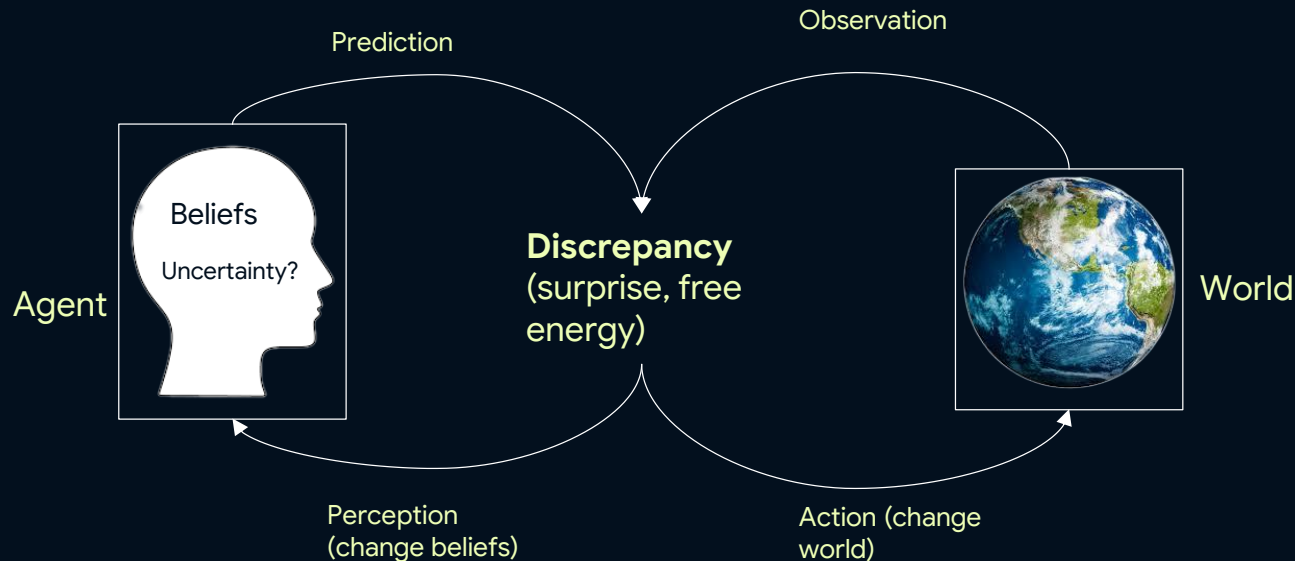


Key challenge: Great diversity of driving scenarios and behavioral aspects that need to be modeled



Need for a generalized, unified modeling framework!

Active inference offers unified framework for road user behavior modeling - both conceptual and computational



- Starting from **first principles** of cognition and behavior in living systems
- **But what are these first principles?**

What is special about living agents?

They make themselves happen!

Cold

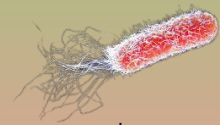


Hot

Food

Tschantz, A., Seth, A.K. and Buckley, C.L. 2020. Learning action-oriented models through active inference. *PLOS Computational Biology*.

Chemotaxis



No food

How do living agents make themselves happen?

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Enactive cognitive science (Maturana and Varela, 1972; Varela et al., 1991; Thompson, 2007)

Constantly **self-produce**, metabolically and behaviorally, to persist as an entity over time - **autopoiesis, sensory-motor autonomy**

Generates and sustains its own activity (autonomy) -> **enacts** (brings forth) **its own meaningful environment (Umwelt)** defining what is **relevant** to the agent given the type of organism it is (sense-making)

Implies **normativity** in interactions with the world and **life-mind continuity**

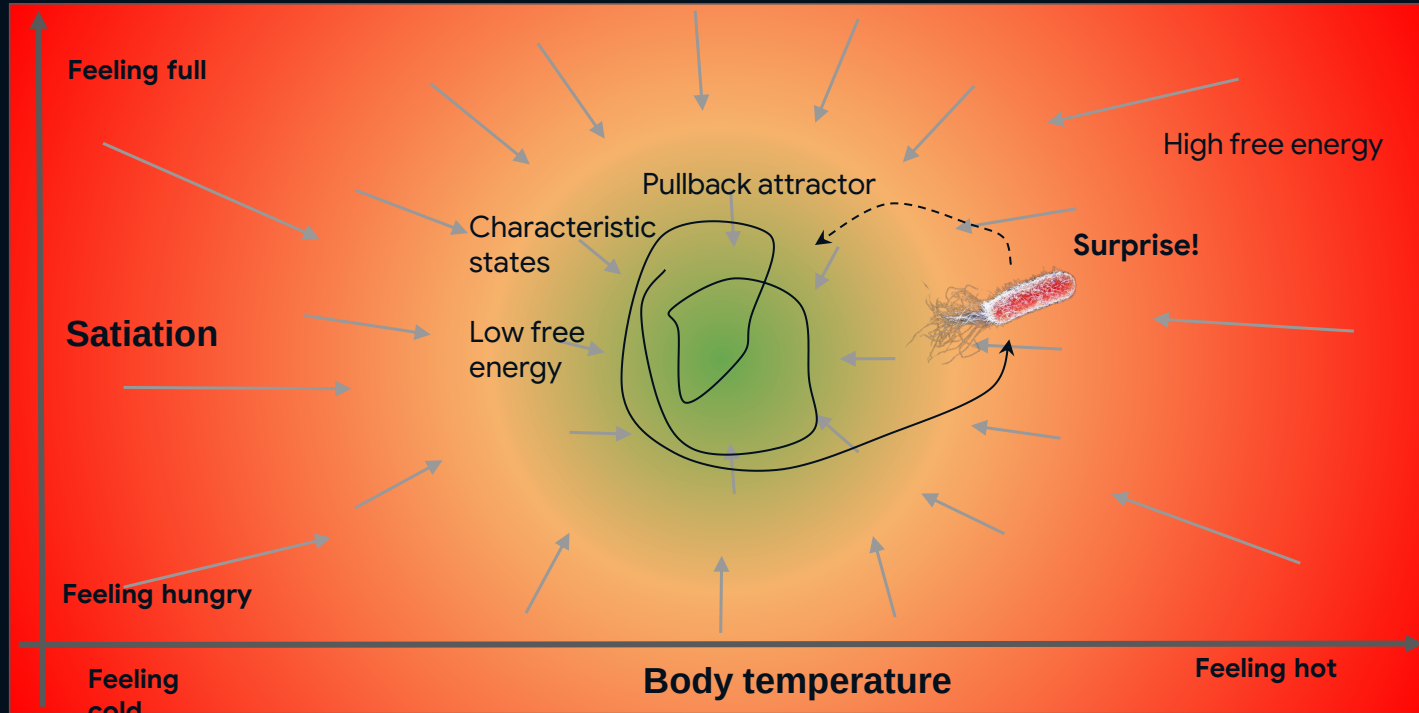
The free energy principle / active inference (Friston, 2010; Parr et al., 2022; Howhy, 2026)

Minimise surprise / free energy to continue existing - a living system **must** minimize free energy to sustain its identity over time

Self evidencing - act to seek observations that provide evidence for their existence (observing being in characteristic, viable, states)

Active inference provides a computational framework for modeling enactive agents - (en)active inference (Kiverstein et al., 2022; Ramstead et al., 2020)

Adaptivity and autonomy as surprise minimization

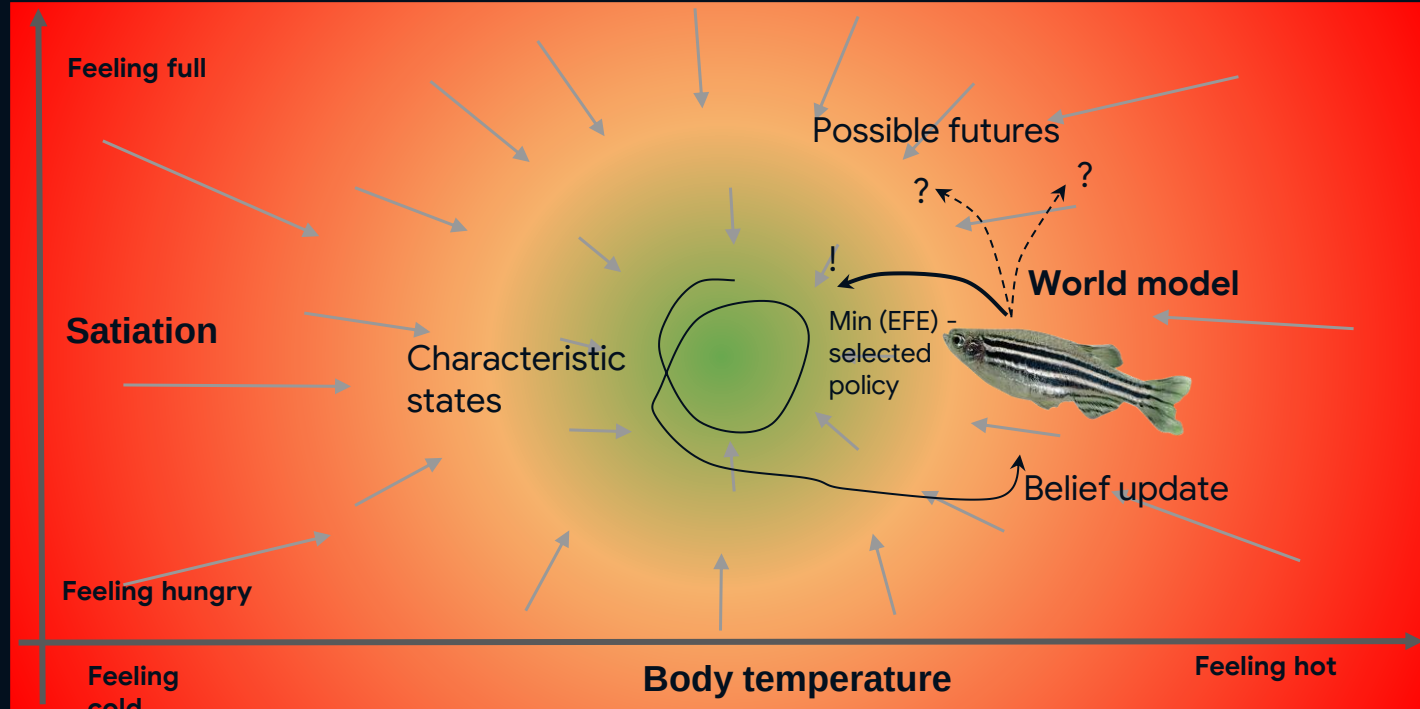


- Living agents expect to be in their characteristic states (and continue existing)
- Their behavior can be understood as minimizing surprise or free energy (an upper bound on surprise)

Agents with brains: Planning based on predicting the consequences of their actions (planning as inference)

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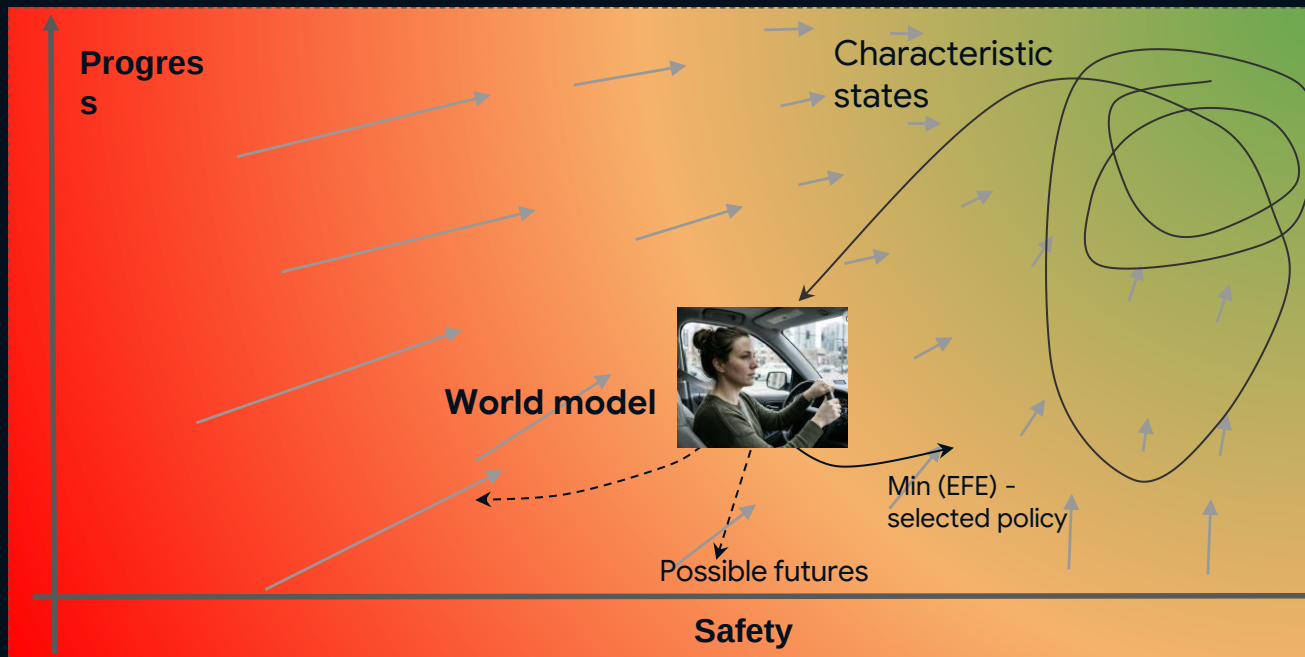
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- **Update beliefs** about the current state based on observations: Minimize **variational free energy** (Bayesian inference)
- **Select action policies** (possible futures) that minimize future surprise: Minimize **expected free energy**

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Generalizing to human drivers



$$\text{Expected free energy (EFE)} = - \underbrace{\text{pragmatic value}}_{\text{Utility (minimize deviation from characteristic states)}} - \underbrace{\text{epistemic value}}_{\text{Resolving uncertainty in beliefs}}$$

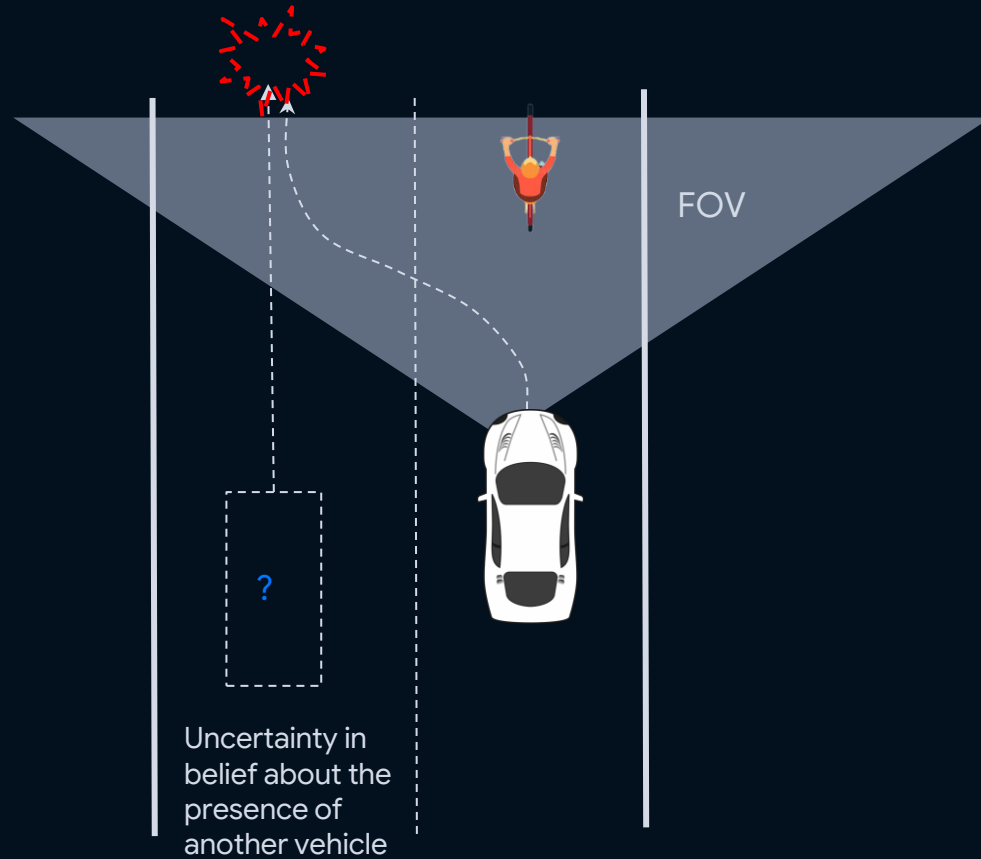
Utility (minimize deviation from characteristic states)

Resolving uncertainty in beliefs

The key notion of a (generative) world model

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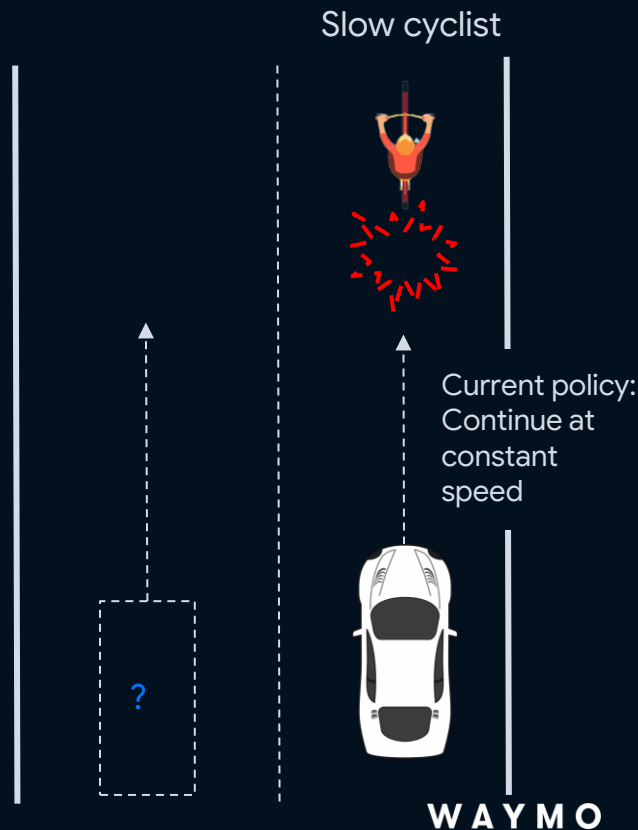


Generates the possible future consequences of a given policy in terms of **what matters** to the particular agent (its Umwelt) -> solves **the relevance problem**

Conceptual human driver model

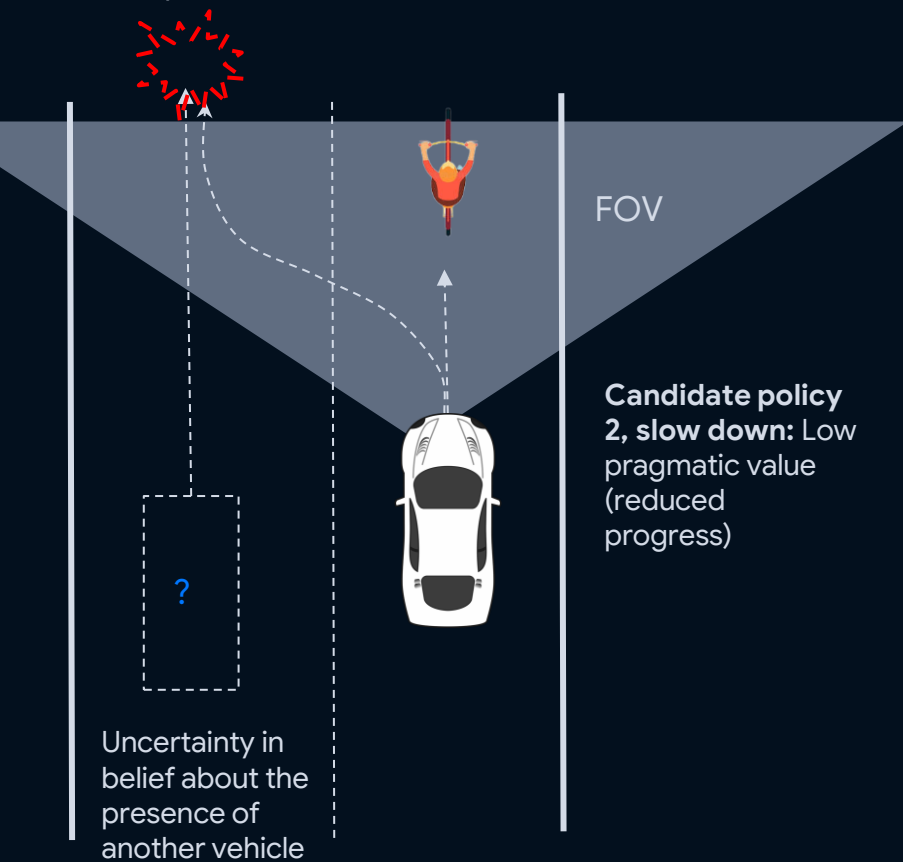
- Characteristic states (preference priors):
 - Make progress
 - Avoid collision and stay on the road
- Drivers maintain a **current policy (plan)** and evaluate its **predicted consequences** given current **beliefs**
- Beliefs may be uncertain due to
 - Partial observability (e.g., occlusions, reduced visibility)
 - Unpredictable behavior of other agents
- If the current policy is “bad” (low pragmatic value), drivers seek alternative policies -> **replanning**
- Policy selection - EFE minimization
 - *High certainty*: Pragmatic actions (move forward to make progress)
 - *Low certainty*: Epistemic actions (visual scanning, communication etc.)

World model



Example: Overtaking a slow cyclist

Candidate policy 1,
overtake: Low pragmatic
value (may lead to crash)



Candidate policy 3,
shoulder check: High epistemic value (resolves uncertainty) -> **unlocks overtaking (Policy 1)**

FOV



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Schumann, J., Engström, J., Johnson, L., O'Kelly, M., Messias, J., Kober, J., Zgonnikov, A. Active inference as a model of collision avoidance behavior in human drivers. *Nature Communications*.

Applying the model to a wide range of scenarios and driver behaviors / states

- Conflict avoidance behavior (Engstrom et al., 2024)
- Collision avoidance behavior (Schumann et al., 2026a)
- Road user interaction and collective behavior (Schumann et al. 2026b)
- Emotion (Schumann et al., 2026c)

In collaboration with our academic partners TU Delft and Texas A&M

Conflict avoidance: Managing uncertainty



Engström J, Wei R, McDonald AD, Garcia A, O'Kelly M and Johnson L (2024) Resolving uncertainty on the fly: modeling adaptive driving behavior as active inference. *Front. Neurobot.* 18:1341750.

frontiers | Frontiers in *Neurobotics*

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EDITED BY
Tom Ziemke,
Linköping University, Sweden

REVIEWED BY
Fabio Tanga,
Centro Ricerche FIAT, Italy
Alice Riebe,
University of Trento, Italy

*CORRESPONDENCE
Johan Engström
✉ jengstrom@waymo.com

Resolving uncertainty on the fly: modeling adaptive driving behavior as active inference

Johan Engström^{1*}, Ran Wei², Anthony D. McDonald³,
Alfredo Garcia², Matthew O'Kelly¹ and Leif Johnson¹

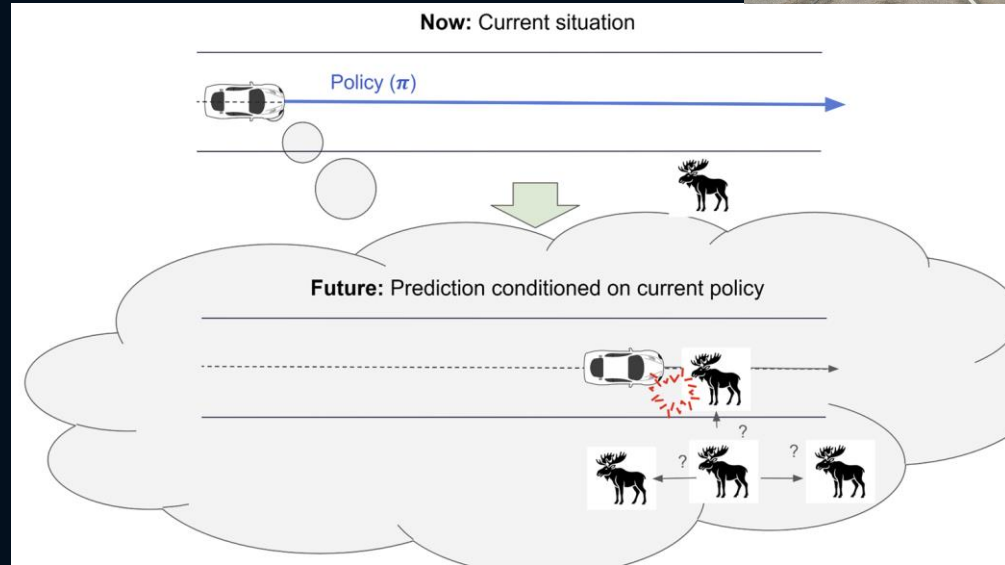
¹Waymo LLC, Mountain View, CA, United States, ²Department of Industrial and Systems Engineering, Texas A&M, College Station, TX, United States, ³Department of Industrial and Systems Engineering, University of Wisconsin-Madison, Madison, WI, United States

YMO

Conceptual interpretation



- Two ways to manage uncertainty
 - **Increase safety margins** (slow down, increase distance) - pragmatic
 - **Resolve the uncertainty** (gain information, look around, communicate) - epistemic
- In reality these are typically intertwined - “resolving uncertainty on the fly” (Engstrom et al. 2024)



Current policy -> *potentially* bad consequence (crash) -> high EFE

Replan -> slow down

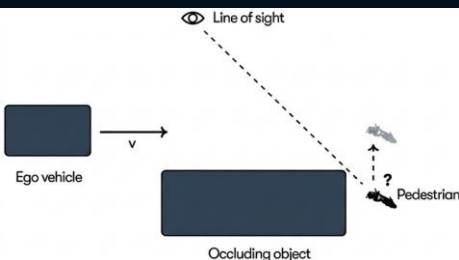
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Simulation example: Occlusion scenario

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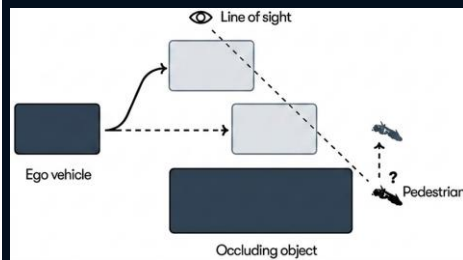
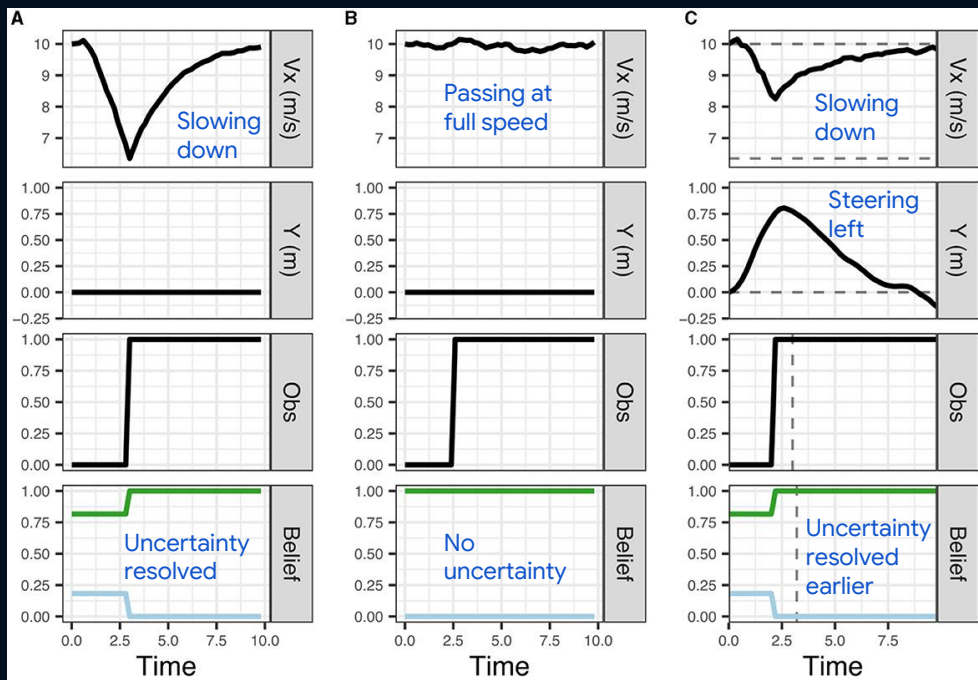
Occlusion scenario



Uncertain if pedestrian is present - ego **can only brake**

Certain no pedestrian is present

Uncertain if pedestrian is present - ego **can brake and steer**



Engström J, Wei R, McDonald AD, Garcia A, O'Kelly M and Johnson L (2024) Resolving uncertainty on the fly: modeling adaptive driving behavior as active inference. *Front. Neurorobot.* 18:1341750.

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Collision avoidance: Urgent replanning to find a way out (an escape affordance)



Schumann, J., Engström, J., Johnson, L., O'Kelly, M., Messias, J., Kober, J., Zgonnikov, A. Active inference as a unified model of collision avoidance behavior in human driver. Nature Communications, <https://doi.org/10.1038/s41467-026-73345-0>

nature communications



Article

<https://doi.org/10.1038/s41467-026-73345-0>

Active inference as a model of collision avoidance behavior in human drivers

Received: 2 June 2025

Accepted: 6 May 2026

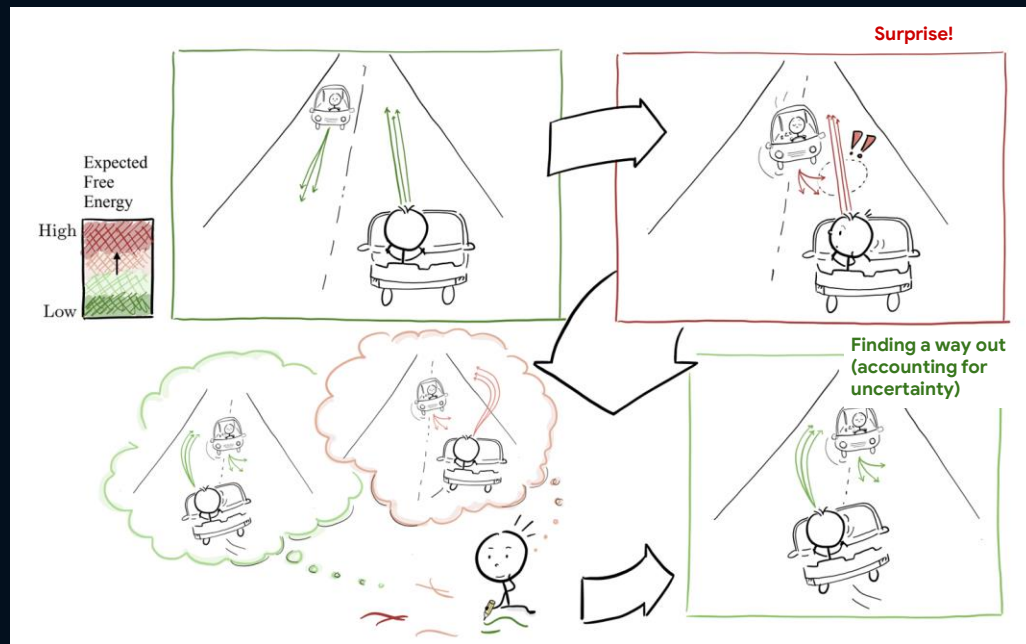
Published online: 10 June 2026

Julian F. Schumann¹, Johan Engström², Leif Johnson², Matthew O'Kelly², Joao Messias², Jens Kober¹ & Arkady Zgonnikov¹

Collision avoidance – involving a rapid threat detection and quick execution of

Conceptual interpretation

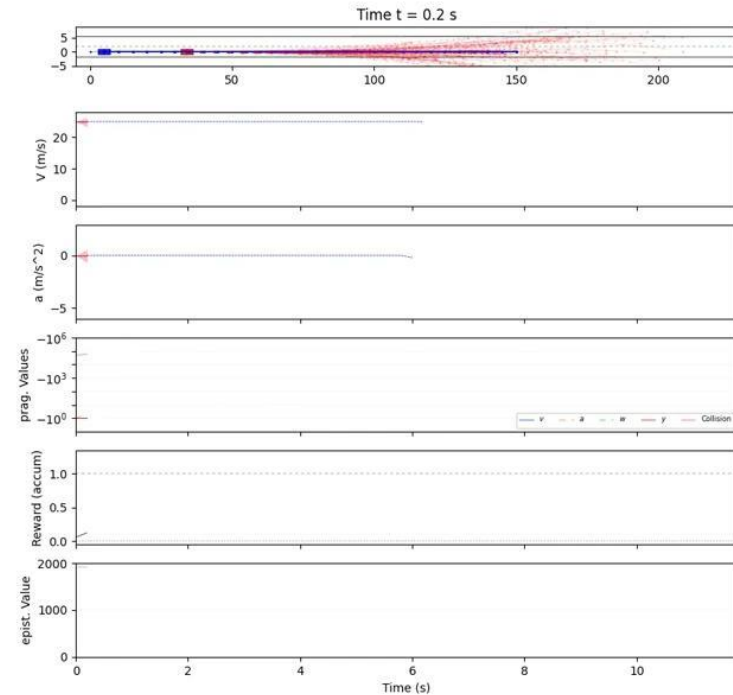
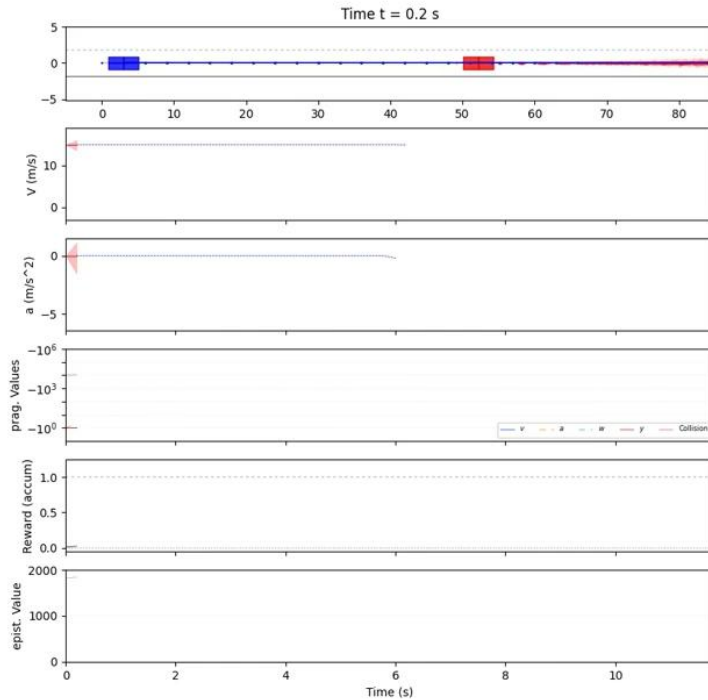
- Rapid development of threat
- Currently policy entails unsafe (uncharacteristic) states -> low EFE (surprise)
- Urgent need to re-plan and find a new policy to avoid collision (escape affordance)
- Find escape affordance by minimizing EFE
- Replanning takes time -> accumulation of evidence -> **response time**
- **Closed loop process** - may need to replan again if conditions change



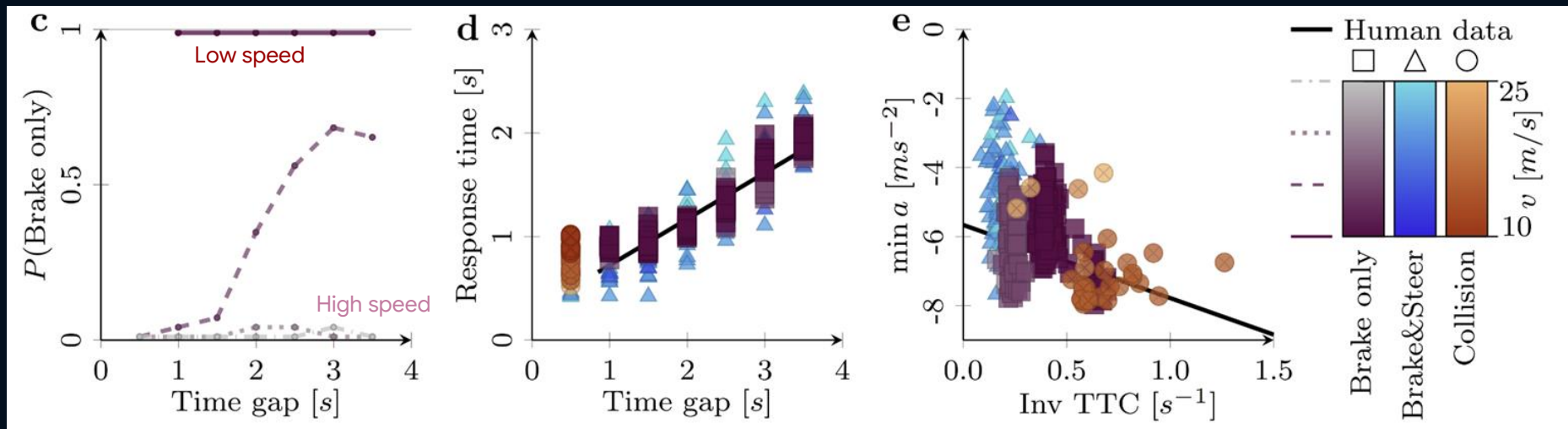
Simulation example: Lead vehicle braking

Low speed

High speed



Comparison with human data

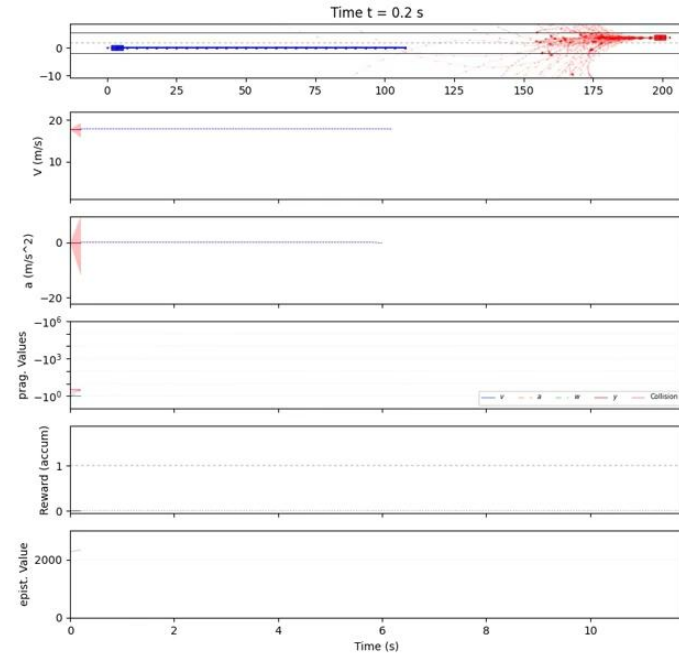
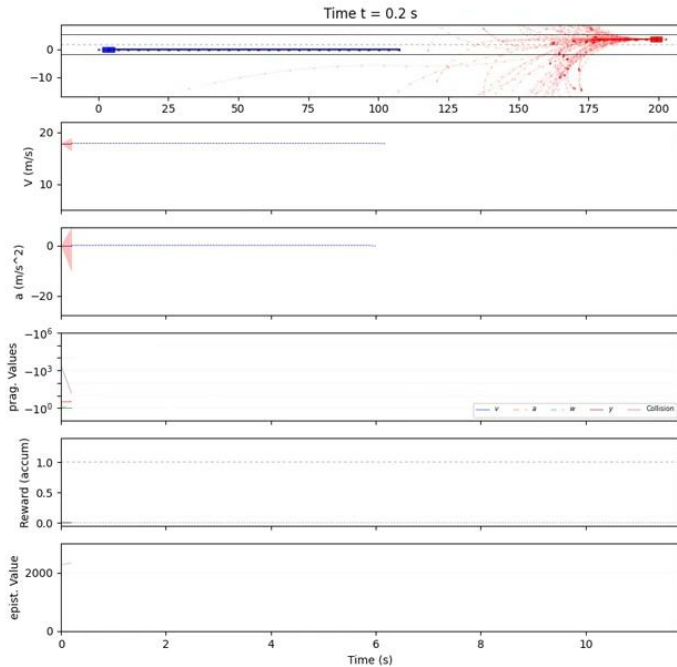


- As the speed increases, probability of brake-only response decreases from 1 to 0 (prediction)
- Brake/steer response times increase with time gap
- Braking intensity increases with increasing situation criticality

Simulation example: Head on conflict

Clear escape affordance to the left (steep incursion)

No escape affordance due to uncertainty (medium incursion)



Comparison with human data



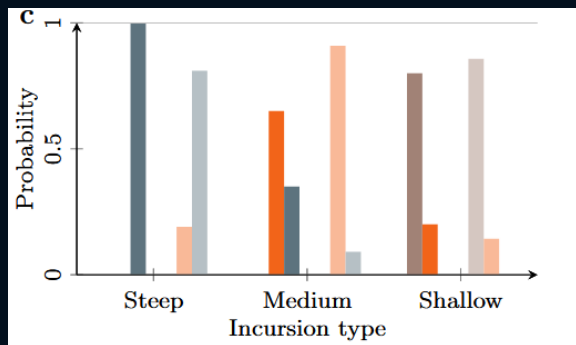
Steep incursion (IL=-0.8)



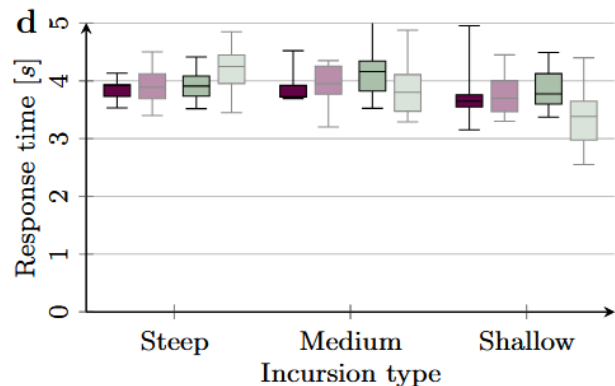
Medium incursion (IL=0)



Shallow incursion (IL=0.9)



Johnson, L., Engström, J., Srinivasan, A., Öztürk, I., & Markkula, G. (2025). *Looking for an out: Affordances, uncertainty and collision avoidance behavior of human drivers* (arXiv:2505.14842).

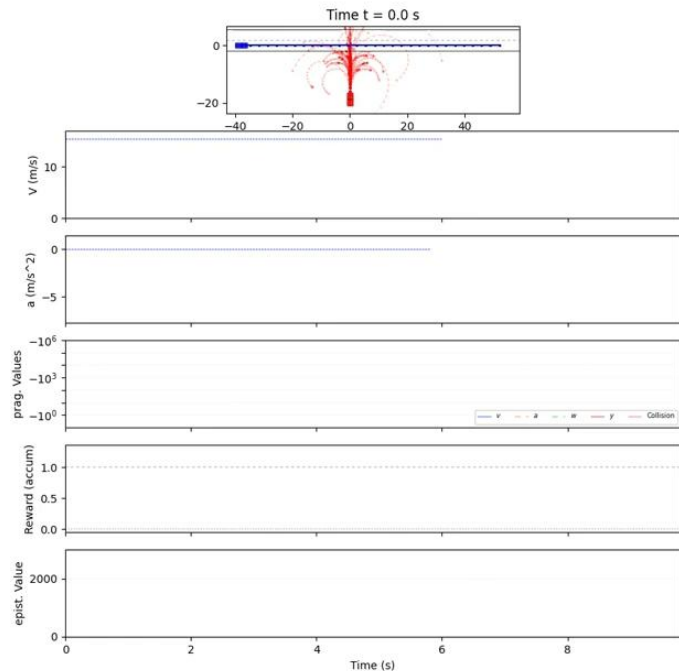
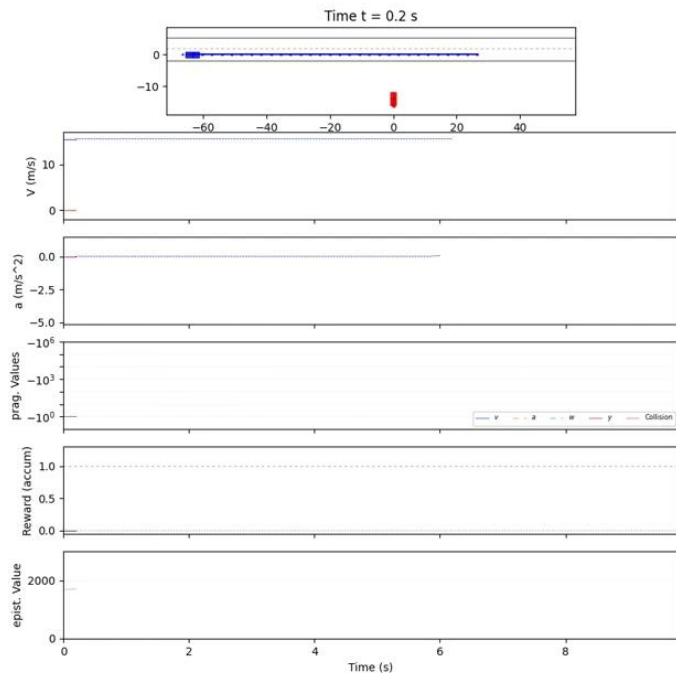


- The model collides in similar situations and with similar frequency as humans
- The model chooses the same evasive maneuvers as humans
- The timing of braking and steering responses is consistent with humans

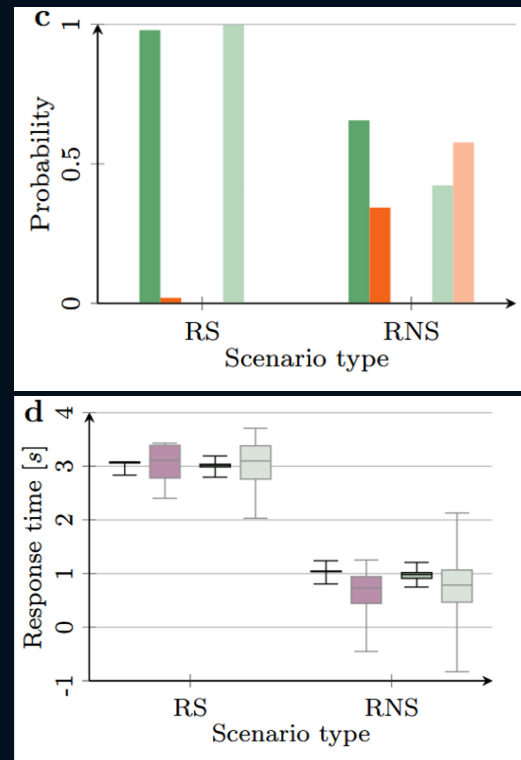
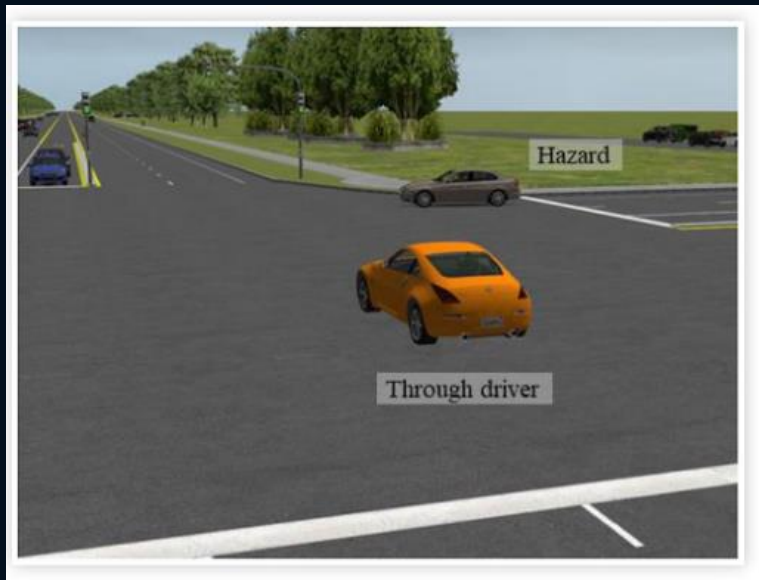
Simulation example: Intersection conflict

Right-turn stopped (RS): Other agent accelerates from stop - less urgent

Right-turn non-stopped (RNS):
Other agent approaches at speed
- more urgent



Comparison with human data



- Model generalized to a completely new, unseen scenario
 - No changes to the model
 - No parameter tuning

Road user interaction: Establishing a shared understanding

Schumann, J.F, Engström, J., Wei, R., Liu, S-Y, Kober, J. and Zgonnikov, A. 2026. Resolving space-sharing conflicts in road user interactions through uncertainty reduction: An active inference-based computational model.

<https://arxiv.org/abs/2604.19838>

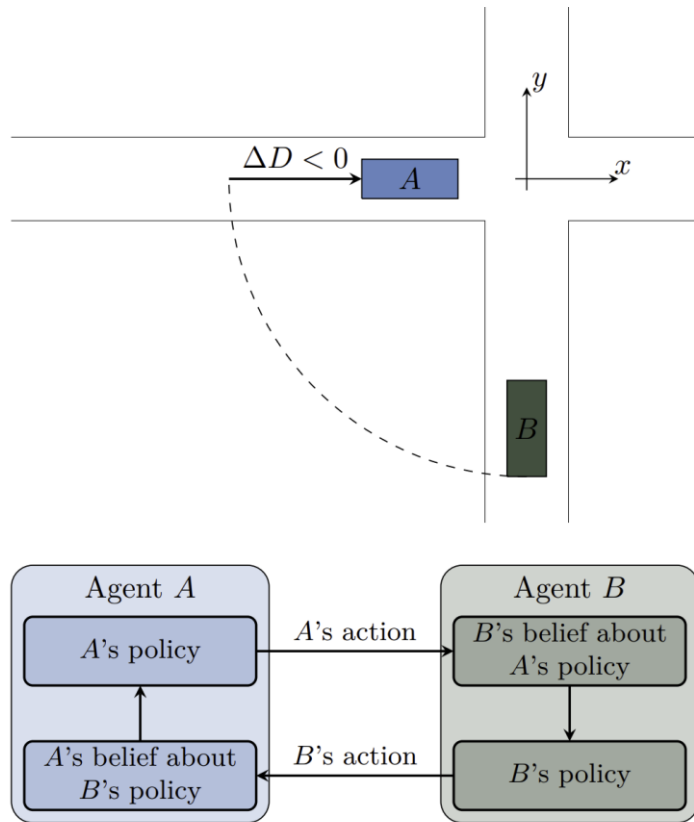


Conceptual interpretation

- Goal of road user interaction is to **resolve space sharing conflicts** (Markkula et al., 2020)
- Requires the establishment of a **shared beliefs** (Pezzulo et al., 2026) or common ground (Clark and Brennan, 1991)
- **Shared beliefs** - agents have the same expectation how the situation will play out - like singing or dancing together (Friston and Frith, 2015)
- Requires **resolving uncertainty in mutual beliefs**
- Two ways to resolve uncertainty
 - Relying on **normative expectations** (e.g., road rules) supported by deontic cues (e.g. traffic signs and signals) - **niche construction**
 - **Explicit communication** (e.g., gesturing) - epistemic action



Is recursive (sophisticated) inference based on theory of mind needed to model road user interactions?

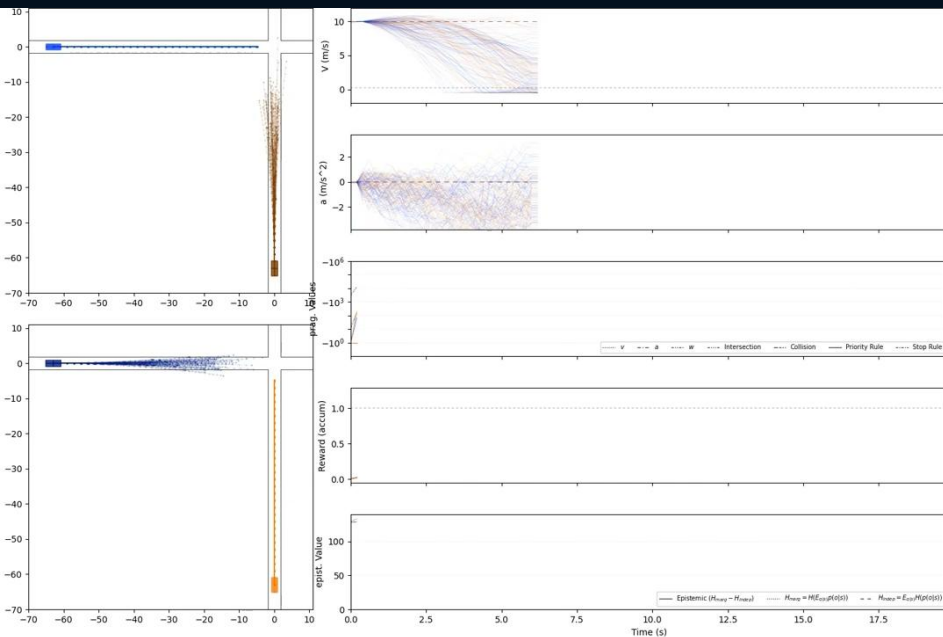


Schumann, J.F, Engström, J., Wei, R., Liu, S-Y, Kober, J. and Zgonnikov, A. 2026. Resolving space-sharing conflicts in road user interactions through uncertainty reduction: An active inference-based computational model. <https://arxiv.org/abs/2604.19838>

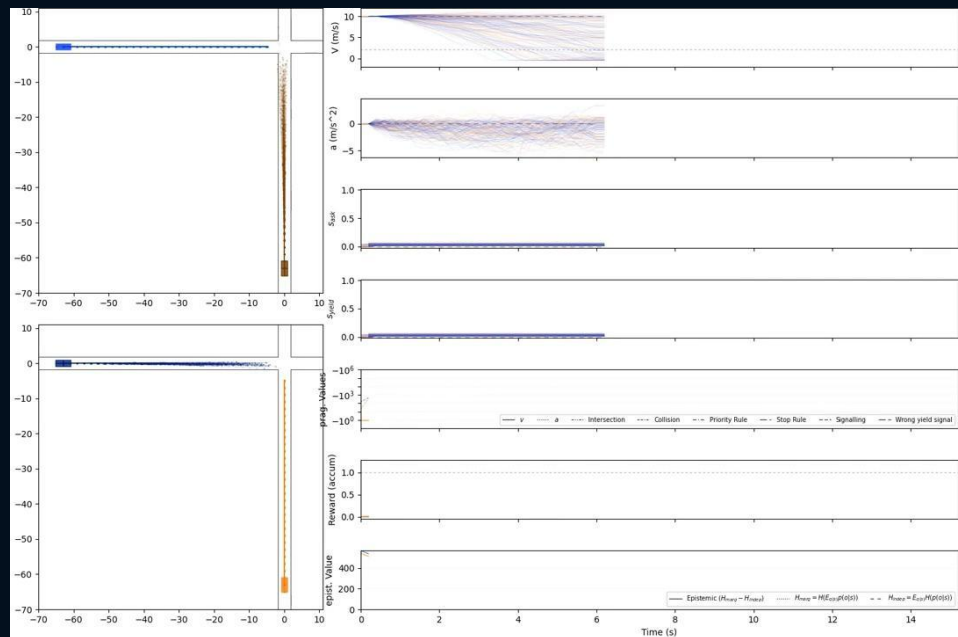
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Simulation examples

Resolution by relying on normative expectations
(stop sign and precedence rule)



Resolution by explicit communication



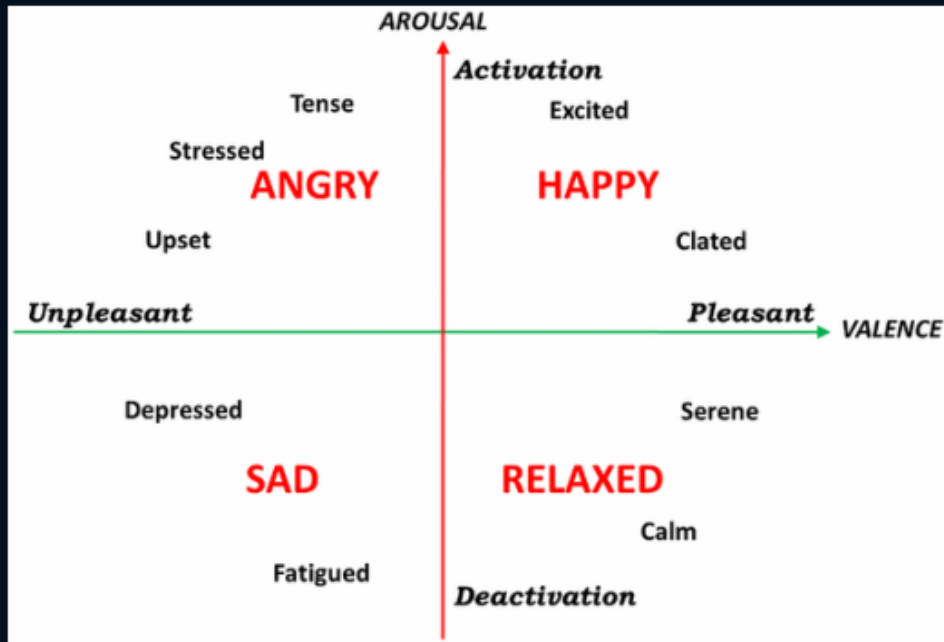
Emotion in driving: Combining valence and arousal



Conceptual interpretation

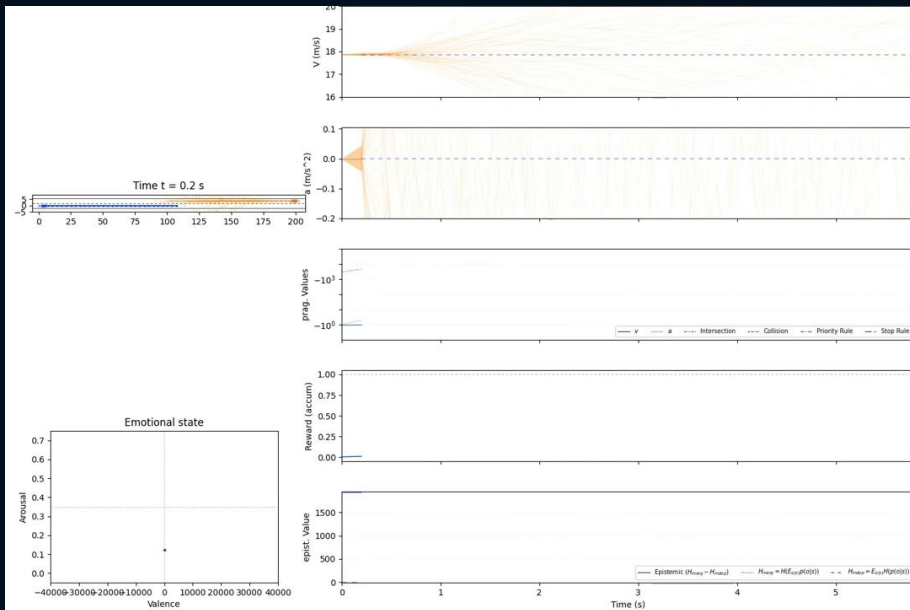
- Active inference interpretation (Pattisapu et al. 2024)
 - Valence - difference between expected and obtained utility (pragmatic value)
 - Arousal - uncertainty (entropy) in posterior beliefs
- Generalizing to dynamic traffic situations
 - Valence - difference between the utility of the current policy and (average of) N previous policies
 - Arousal - entropy in posterior beliefs about future collisions

The circumplex model of emotion (Russel, 1980)

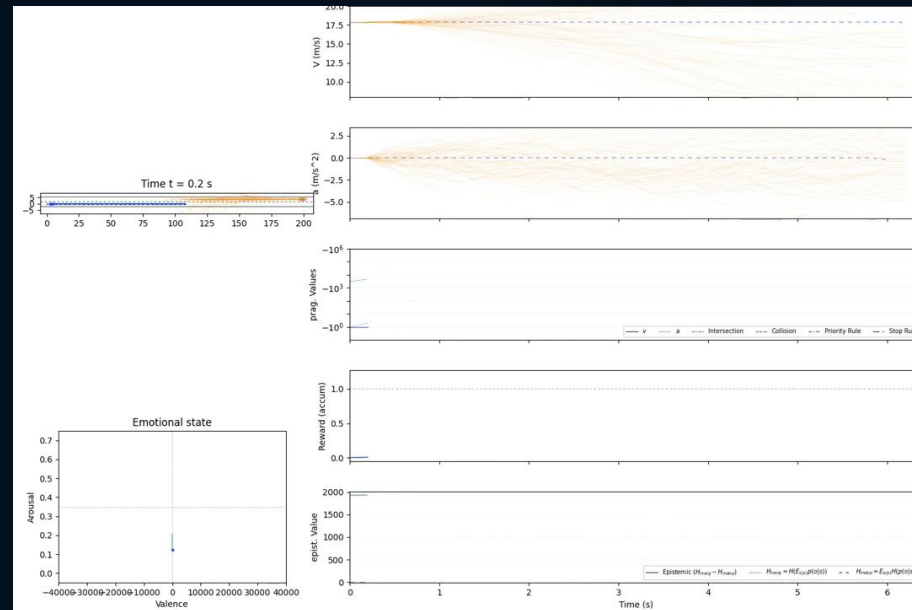


Simulation example: Head-on scenario

No incursion -> minor emotional reaction



Incursion -> major emotional reaction



Productionizing the model: The Waymo Reference Driver (ReD)

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Key takeaways

- Computational models of human drivers are of key importance for ADS development and evaluation - e.g., as a behavioral benchmark for the Waymo driver
- Models need to be strongly generalizable to cover wide ranges of scenarios and driving behaviors
- Active inference framework offers a generalizable framework - both conceptually and for computational modeling
- Shown here that a single driver model can...
 - ... reproduce human behavior in three different collision avoidance scenario types (F2R, head-on, intersection)
 - ... account for a range of behaviors / states (conflict avoidance, collision avoidance, interaction, emotion)

Areas of future work

- Ongoing
 - Visual behavior and crash causation
 - Human takeover behavior
 - Parameter learning/tuning
 - Integration of ML-based world models
 - Modeling collective behavior and normative expectations (with Dr. Axel Constant)
- Future topics
 - More abstract state representations
 - Give emotions a more causal role in the model
 - Attention, automaticity and cognitive control
 - ...

TU Delft



Julian Schumann



Jens Kober



Arkady Zgonnikov

Texas A&M



Tony McDonald



Ran Wei



Alfredo Garcia

Waymo



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